Linux Log File Analysis, Automation, and SIEM Visualization

Resources: https://github.com/logpai/loghub/blob/master/Linux/Linux_2k.log

1. Manual Log Files

Objective: Analyze computer logs to identify suspicious login activity.

Tools: Basic knowledge of authentication and SSH, Excel, Sample log file(resource above)

Date/Time	Source IP / Host	Event Type	Target User	Notes
6/14/26 15:16	218.188.2.4	Authentication Failure	Unknown	User unknown attempt
6/14/26 15:16	N/A	Check Pass (User Unknown)	Unknown	User does not exist
6/14/26 15:16	218.188.2.4	Authentication Failure	Unknown	User unknown attempt
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
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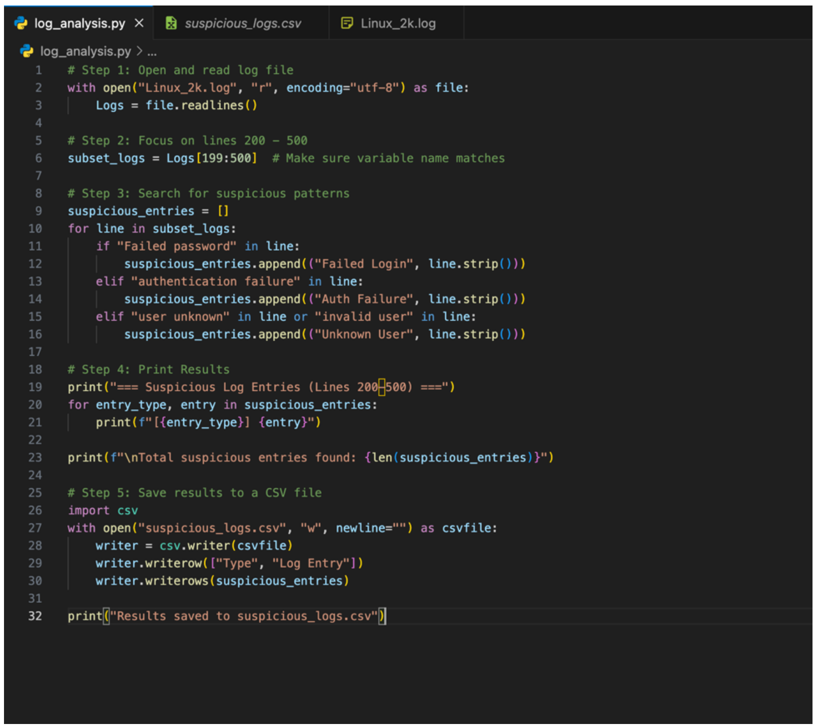
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 2:04	220-135-151-1.hinet-ip.hinet.net	Authentication Failure	root	brute force
6/15/26 4:06	Localhost	Session Opened	cyrus	Root switched to user cyrus
6/15/26 4:06	Localhost	Session Closed	cyrus	Session ended
6/15/26 4:06	Localhost	Logrotate Alert	N/A	logrotate exited abnormally [1]
6/15/26 4:12	Localhost	Session Opened	news	Root switched to user news
6/15/26 4:12	Localhost	Session Closed	news	Session ended
6/15/26 12:12	N/A	Check Pass (User Unknown)	Unknown	User does not exist
6/15/26 12:12	218.188.2.4	Authentication Failure	Unknown	User unknown attempt

The logs indicate several failed login attempts from the IP addresses 218.188.2.4 and 220.135.151.1, targeting mainly the root account. This could mean that malicious actors are trying a brute force attack. All those failures mean it is an automated attack or probing. The unknown users also imply that the attackers are trying various usernames. Additionally the alert can suggest a system issue that requires further investigation.

2. Automating Log Analysis

Objective: Extend the manually review by writing a simple Python Script in VS code to automatically scan Linux Log files.

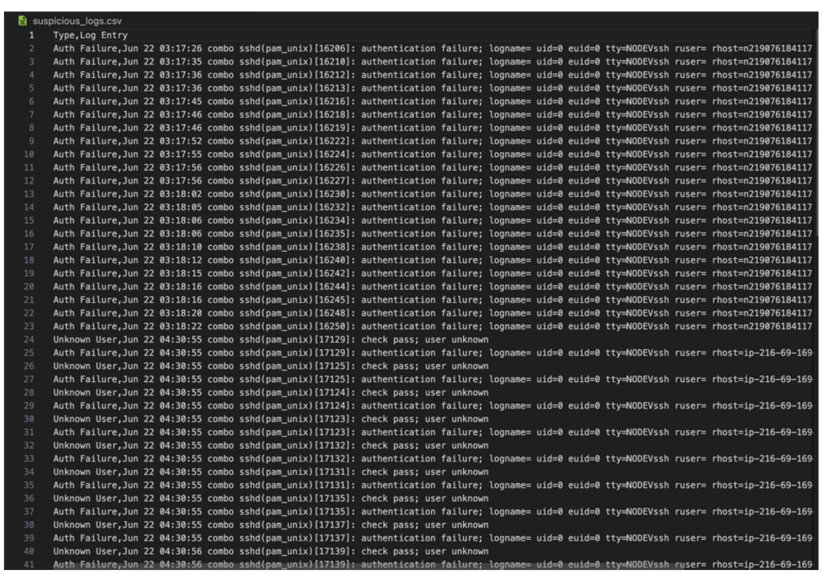
Tools: VS Code, Python 3.x, Basic Knowledge of Python, Samples Log File



log_analysis.py × | suspicious_logs.csv | Linux_2k.log

log_analysis.py > ...

```python
# Step 1: Open and read log file
with open("Linux_2k.log", "r", encoding="utf-8") as file:
    Logs = file.readlines()

# Step 2: Focus on lines 200 - 500
subset_logs = Logs[199:500]  # Make sure variable name matches

# Step 3: Search for suspicious patterns
suspicious_entries = []
for line in subset_logs:
    if "Failed password" in line:
        suspicious_entries.append(("Failed Login", line.strip()))
    elif "authentication failure" in line:
        suspicious_entries.append(("Auth Failure", line.strip()))
    elif "user unknown" in line or "invalid user" in line:
        suspicious_entries.append(("Unknown User", line.strip()))

# Step 4: Print Results
print("=== Suspicious Log Entries (Lines 200–500) ===")
for entry_type, entry in suspicious_entries:
    print(f"[{entry_type}] {entry}")

print(f"\nTotal suspicious entries found: {len(suspicious_entries)}")

# Step 5: Save results to a CSV file
import csv
with open("suspicious_logs.csv", "w", newline="") as csvfile:
    writer = csv.writer(csvfile)
    writer.writerow(["Type", "Log Entry"])
    writer.writerows(suspicious_entries)

print("Results saved to suspicious_logs.csv")
```



suspicious_logs.csv

```
Type,Log Entry
Auth Failure,Jun 22 03:17:26 combo sshd(pam_unix)[16206]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:35 combo sshd(pam_unix)[16210]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:36 combo sshd(pam_unix)[16212]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:36 combo sshd(pam_unix)[16213]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:45 combo sshd(pam_unix)[16216]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:46 combo sshd(pam_unix)[16218]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:46 combo sshd(pam_unix)[16219]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:52 combo sshd(pam_unix)[16222]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:55 combo sshd(pam_unix)[16224]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:56 combo sshd(pam_unix)[16226]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:17:56 combo sshd(pam_unix)[16227]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:02 combo sshd(pam_unix)[16230]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:05 combo sshd(pam_unix)[16232]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:06 combo sshd(pam_unix)[16234]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:06 combo sshd(pam_unix)[16235]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:10 combo sshd(pam_unix)[16238]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:12 combo sshd(pam_unix)[16240]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:15 combo sshd(pam_unix)[16242]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:16 combo sshd(pam_unix)[16244]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:16 combo sshd(pam_unix)[16245]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:20 combo sshd(pam_unix)[16248]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Auth Failure,Jun 22 03:18:22 combo sshd(pam_unix)[16250]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=n219076184117
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17129]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17129]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17125]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17125]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17124]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17124]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17123]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17123]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17132]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17132]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17131]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17131]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17135]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17135]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:55 combo sshd(pam_unix)[17137]: check pass; user unknown
Auth Failure,Jun 22 04:30:55 combo sshd(pam_unix)[17137]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
Unknown User,Jun 22 04:30:56 combo sshd(pam_unix)[17139]: check pass; user unknown
Auth Failure,Jun 22 04:30:56 combo sshd(pam_unix)[17139]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=ip-216-69-169-
```

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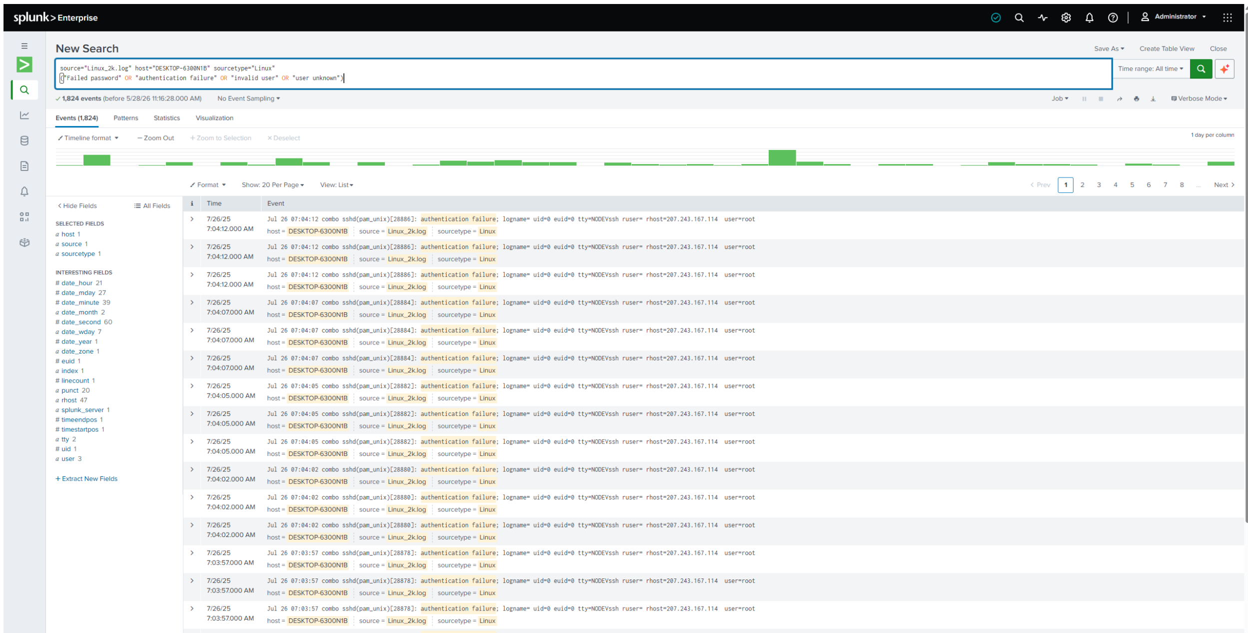
This Python script opens the Linux_2k.log file and reads the lines into a log. It then takes lines 200 to 500 and reads and searches for specific keywords and each line that contains those keywords is put into the suspicious_entries list. So the script puts the falgged entries into a list and aslo gives the total number of entries. It then saves it into a new file so that it is easy t oread or export.

This code automates the detection of failed logins, authentication failures, and unknown users. It also generates a CSV report.

3. Log Analysis and Visualization with Splunk

Objective: Use Splunk(SIEM Tool) to build on the manual and automated log analysis. Ingest, query, and visualize Linux log data identify suspicious activities at scale.

Tools: Access to Splunk(Enterprise Edition), log sample



In Spunk a dataset was added containing the log sample file. With the search shown above it filtered the logs to show only the entries containing:

- Repeated failed logins
- Invalid or unwon users
- Authentication Failures

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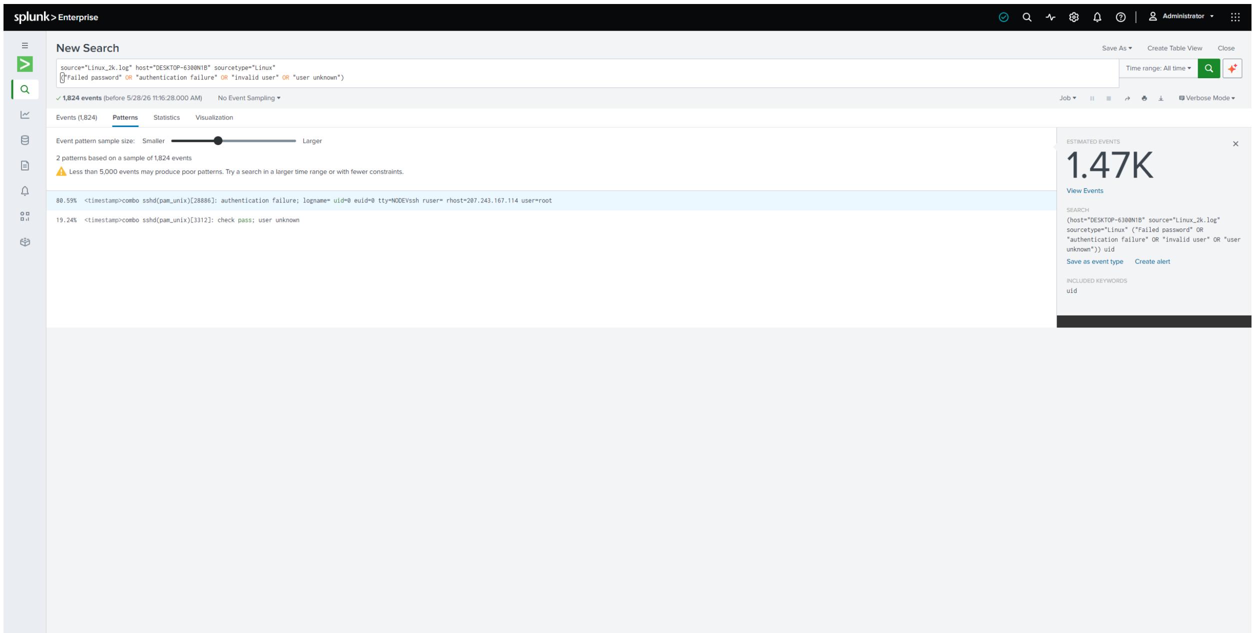
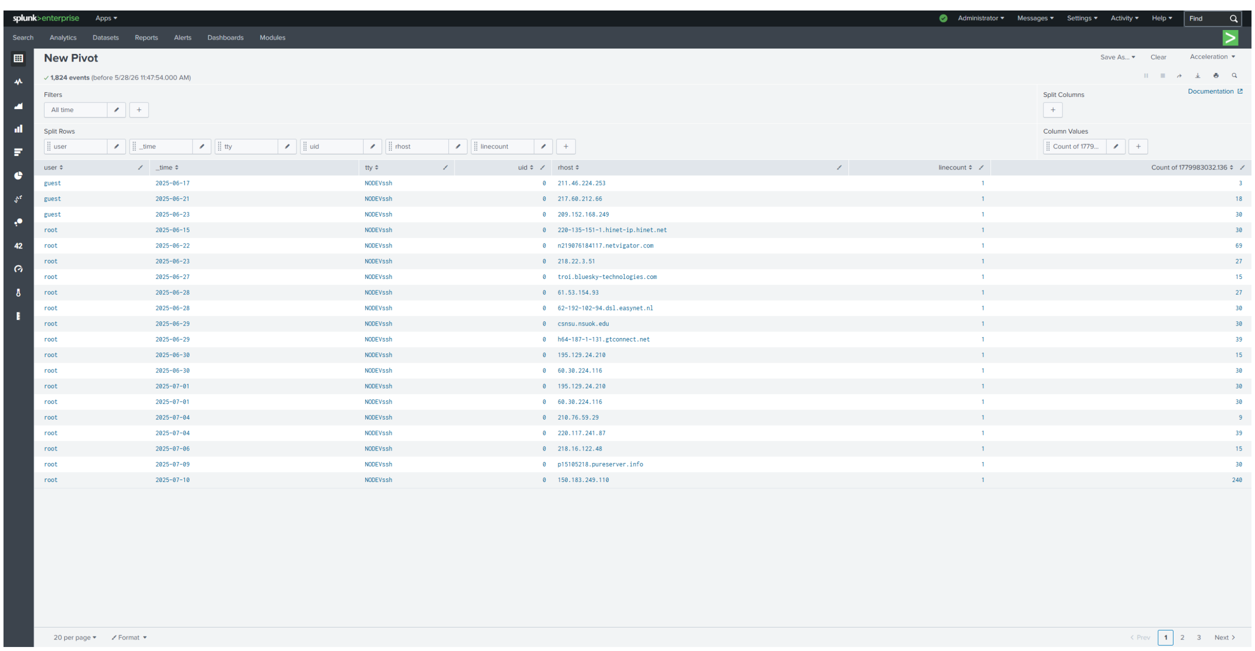
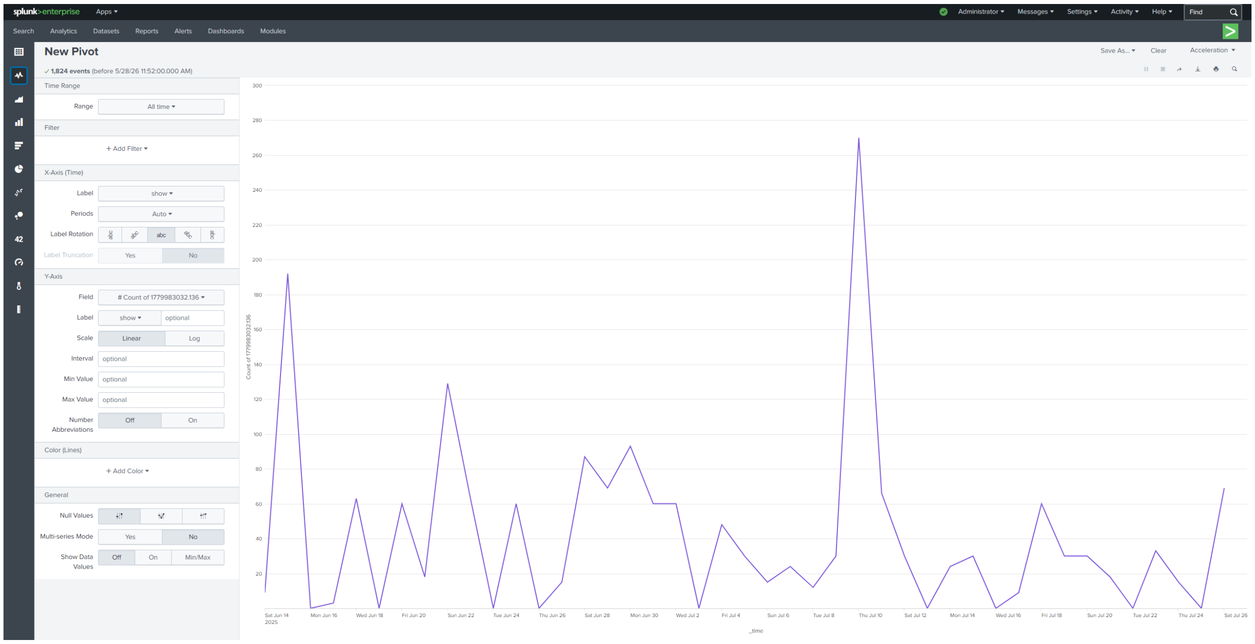


One can switch to *Patterns View* which can be used to confirm repetitive behavior.

One can see that “Failed Password for root” appears repeatedly, this confirms that multiples source IP generated repeated failures using the same target.



Similarly one can switch to *Statistics View* to have a more in depth look at the data. It helps to break down the data to better see the scale and focus of the attack.



Finally one can switch to the *Visualization View*. It helps to better view the timeline of the attacks. It helps to better identify patterns and potential security incidents. Each spike represent multiple events in short intervals. The activity is time based and comes in bursts, likely indicating brute force attempts. Attackers are probably using scripts.

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Summary

This project focused on analyzing Linux authentication logs to identify suspicious login activity and demonstrate different approaches to security monitoring and incident detection. The project was divided into three phases: manual log analysis, automated log analysis using Python, and visualization using Splunk SIEM.

During the manual analysis phase, Linux log files were reviewed to identify failed login attempts, authentication failures, and invalid user activity. Multiple failed login attempts targeting the root account were observed from IP addresses such as 218.188.2.4 and 220.135.151.1. The repeated failures and use of unknown usernames suggested possible brute-force or automated attack activity.

The second phase expanded the analysis through automation using Python in Visual Studio Code. A custom script was developed to scan Linux log files and detect suspicious entries based on keywords such as failed passwords, invalid users, and authentication failures. The script extracted suspicious events, counted the total number of detections, and exported the results into a CSV report for easier review and documentation. This demonstrated the ability to automate repetitive security analysis tasks and improve efficiency when handling larger datasets.

The final phase used Splunk Enterprise to ingest, search, and visualize Linux log data at scale. Splunk searches were used to filter suspicious authentication events and identify repeated attack patterns. Pattern View confirmed repetitive failed password attempts against privileged accounts, while Statistics and Visualization views helped analyze attack frequency, source IP activity, and time-based spikes. The visualization showed bursts of repeated login attempts that are commonly associated with scripted brute-force attacks.

Overall, this project demonstrated foundational cybersecurity and SOC analyst skills including:

* Log analysis and threat detection

Linux Log File Analysis, Automation, and SIEM Visualization

* Linux authentication monitoring
* Identifying brute-force attack behavior
* Python scripting for security automation
* SIEM usage with Splunk
* Data filtering, reporting, and visualization
* Basic incident investigation techniques